

Thabiso Shongwe

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<https://www.linkedin.com/in/thabiso-shongwe/> | <https://github.com/ClaydeThabiso>

Summary

Final-year BSc Computer Science and Informatics student at the University of Johannesburg with hands-on experience in **C++**, **Python**, **Java**, **ASP.NET**, **SQL**, and **full-stack web development**. Skilled in **designing, developing, and deploying efficient, scalable applications**, mentoring peers, and solving complex programming challenges. Passionate about leveraging technology to create real-world solutions while continuously expanding technical expertise.

Education

University of Johannesburg, Bachelor of Science in Computer Science and Informatics Expected 12/2026

- JBS Grit lab 2023 graduate

Experience

Computer Science Tutor, Academy of Computer Science and Software Engineering (ACSSE), University of Johannesburg February 2026 – Present

- Provide one-on-one and small-group tutoring in core Computer Science modules
- Simplify complex programming and theoretical concepts into clear, practical explanations
- Assist students with assignment preparation, exam readiness, and coding best practices

Grit lab intern, JBS(Johannesburg Business School) October 2023 – Present

- Engaged in real-world business and technology challenges, collaborating with diverse teams to develop scalable solutions.
- Working on data-driven projects that applied AI and advanced technologies to solve complex problems.
- Developed critical thinking, resilience, and adaptability through rigorous training and mentorship.

Projects

Banking Web Application [github.com/BankingWebApp](https://github.com/ClaydeThabiso/BankingWebApp)

- Developed a banking web application to stimulate basic banking functionality allowing users to manage accounts and transactions.It provides a user-friendly interface for performing CRUD operations on customer information,account details and transaction records.
- Tools Used: C#, ASP.NET, SQL, HTML5, CSS3, Bootstrap 5

iSave-E-commerce-Shop [github.com/iSave](https://github.com/ClaydeThabiso/iSave)

- iSave is a smart web-based platform offering energy-saving products and technologies to homes and small businesses. In an age where electricity costs are rising and climate change is a serious concern, this platform makes it easy for people to shop for energy-efficient products and get data-driven insights on how much energy and money they'll save over time.
- Tools Used: C#, ASP.NET, SQL, HTML5, CSS3, Bootstrap, JavaScript

Disaster Relief Routing System [github.com/DisasterRouting](https://github.com/ClaydeThabiso/DisasterRouting)

- Developed a Disaster Relief Routing System that transforms map images into navigable graphs and computes optimal safe paths using Dijkstra's algorithm with risk-aware weighting.
- Key Features: Image-to-Graph conversion, terrain classification, custom Binary Heap Priority Queue, interactive JavaFX GUI with real-time path visualization.
- Tools: Java, JavaFX, Graph Algorithms, HBPriorityQueue, Map, PixelReader

BridgeBox

[github.com/BridgeBox](https://github.com/ClaydeThabiso/BridgeBox)

- BridgeBox is a modern, gamified personal finance manager. It helps users manage their money by organizing funds into customizable “boxes” while making saving and spending fun through gamification elements like quests, rewards, and XP leveling.
- Tools Used: HTML5, CSS3, JavaScript, Bootstrap 5

VendorConnect

[github.com/VendorConnect](https://github.com/ClaydeThabiso/VendorConnect)

- VendorConnect is a web-based platform designed to streamline the connection between vendors and event organizers. The system supports vendor registration, event applications, approval workflows, notifications, and administrative dashboards. Built using ASP.NET (C#) with a structured SQL Server database and Bootstrap for the frontend, the platform emphasizes secure data handling, role-based access control, and efficient tracking of vendor activities.
- Tools Used: C#, ASP.NET, SQL, HTML5, CSS3, Bootstrap, JavaScript

JBS 3D Unity lab

- Developed a unity simulation of the JBS grit lab
- Tools Used: UNITY 3D

More projects in GitHub: <https://github.com/ClaydeThabiso>

Technologies

Languages: C++, Python, Java, C#, SQL, VB.NET

Web Development: HTML, CSS, Bootstrap, ASP.NET, JavaScript

Technologies: .NET, Microsoft SQL Server, Unity, JavaFx, Overleaf, Git

Software Development: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Engineering Principles, MVC Architecture

References

Prof Abejide Ade-Ibijola (JBS-Grit lab)

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